Ruike Lin

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Education

University of Wisconsin-Madison Madison, WI
Master of Science in Industrial Engineering Jan. 2026 - Present
Relevant Coursework: Human Factors Engineering, Industrial Supply Chain Management, Machine Learning Practices

University of Wisconsin-Madison Madison, WI
Bachelor of Science in Information Science / Data Science (Minor) Sep. 2023 - Dec. 2025
Cumulative GPA: 3.50/4.00 ;
Relevant Coursework: Interaction Design, Data Science Models, Generative AI, Big Data and Artificial Intelligence

Skills

- Data Analytics & Visualization: R, Python, Microsoft Excel, ggplot2, matplotlib, Tableau
- Web Development: HTML, CSS, JavaScript, Bootstrap
- Development Tools: Jupyter Notebook, Visual Studio Code, GitHub, Figma
- Statistical Analysis: ANOVA, t-test, Regression Modeling, Logistic Regression

Project Experience

Cooksy: Smart Cooking Time Assistant App Product Manager Sep. 2025 - Dec. 2025
- Conducted user research through in-depth interviews, focus groups, and contextual inquiry to identify students' pain points in cooking: time management, ingredient selection, and cleaning
- Based on identified shortcomings in competitor products, defined core features: recipe entry, multi-tag recipe search, cooking progress timers, simultaneous input of multiple cooking tasks with intelligent step planning, photo sharing, community interaction, personal profile, and recipe collection. Developed user flowcharts and PRD
- Created 10 pages of wireframes and high-fidelity prototypes in Figma, applying consistent design guidelines and ensuring color visual accessibility, delivered interactive prototypes within the project timeline, featuring the home page, recipe recommender, filters, and personal space

Study of the Impact of Pet Interaction on Emotional Well-being Sep. 2025 - Dec. 2025
- To understand the effect of pet interaction on human mood, recorded interactions with a pet dog and corresponding emotional responses from participants three times daily over 14 days, entered data into Excel sheets and transformed these records into structured data for subsequent analysis using emotional keywords and a simple scoring system
- Analyzed the impact of factors such as the type, duration, and environment of human-pet interaction on emotions, created hierarchical views based on the data tables and generated a presentation file using R Studio
- Data analysis revealed that positive interactions with pets significantly enhance individuals' subsequent focus on work and life. Produced a research paper; the course received a score of 90/100

Google Teachable Machine Facial Expression Recognition Project: Smile Detection Jan. 2024 - May 2024
- Developed a facial expression classifier based on Teachable Machine and JavaScript, achieving real-time detection within seconds. Tested the product's functionality with 20 users
- Built a smile detection web interface using p5.js and ml5.js, integrating the camera with the Teachable Machine model to display real-time classification results and corresponding emojis
- Investigated bias in facial recognition by testing how lighting and skin tone affect model accuracy, Identified accuracy disparities for facial expression recognition across different skin tones in the database, implemented backend measures to balance accuracy across skin tones to mitigate potential product risks

Research Experience

Gravity Spy: Gravitational Wave Data Annotation & Modeling Analysis Sep. 2025 - Jan. 2026
- Optimized data cleaning and annotation workflows for the Gravity Spy gravitational wave project, a collaborative research initiative involving Northwestern University, Syracuse University, and CSUF
- Extracted valid variables from Excel data containing 1.8 million+ user-annotated images; used R Studio to merge datasets and generate a new consolidated document
- To evaluate and improve image classification performance of the existing dataset, compared consistency between human recognition results and machine learning model identifications, discovering significantly lower ML accuracy for high-difficulty images. Implemented difficulty stratification, prioritizing manual review for high-difficulty images
- Investigated behavioral changes and capability improvement trends in users during continuous annotation tasks, providing recommendations for optimizing new-user onboarding, task sequencing, and user retention strategies

Zooniverse Product Optimization Analysis Project (Dr. Corey B Jackson) May. 2024 - Jan. 2026
- Assisted product optimizing for Zooniverse, a non-profit citizen science platform funded by NASA with 2 million users, core function is to provide recruitment portals for citizen science projects and facilitate multi-faceted support
- Processed over 6,000 survey responses from NASA-funded citizen science projects, analyzed correlations between demographic variables and participation levels, identifying disparities in engagement based on community demographics. Found that the core user group consisted of individuals aged 60-65, white, with some college education
- Delivered a full presentation in English at the IDS conference with approximately 40 attendees, explaining the correlation between user participation and value orientation. Contributed as a co-author to the results section, formatted the manuscript on the Overleaf platform; the paper was included in the SSRN database